

Dual-track retraining loop

Weight W

1.2 .3 .9 .1

Soft mask m

.80 .32 .74 .20



Forward and weight update

$\mathcal{L}_{\text{task}}$ $\mathcal{L}_{\text{KL}}$ $\mathcal{L}_{\text{mid}}$ SLoRB B

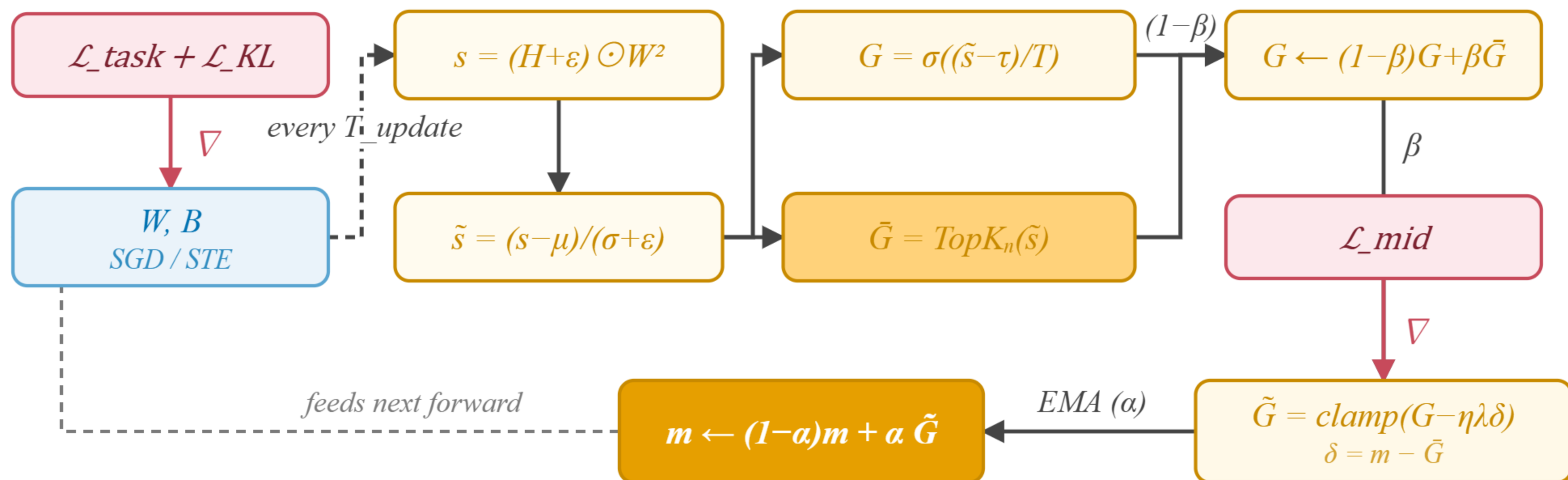
Sparse forward pass
 $\hat{Y} = X(\dot{W}^T + P^T B^T)$

Hessian-guided mask update

δ G $\bar{G}$ $\tilde{G}$

Score function
 $s = (H + \varepsilon) \odot W^2$

Per-iteration update flow



Heating



.55 .50 .53 .48

.78 .25 .74 .18

.96 .06 .94 .04

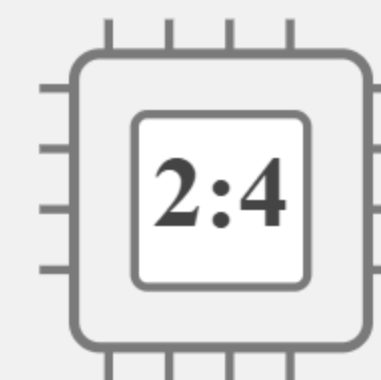
Quenching



Project soft mask to
exact 2:4.

Freeze mask and
finetune weights.

Deployable 2:4 Sparse LLM



1.2 .3 .9 .1

1 0 1 0

1.2 0 .9 0